

Generative AI and Large Language Model Foundations

Controlled Lesson for Production Unit PU B04 C01

Concept	Operational meaning
Book	Book 4 Generative AI and Large Language Models
Lesson	Lesson 1 Generative AI and LLM Foundations
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Audience	Technical practitioners, analysts, evaluators, managers and executive learners

This lesson explains what generative language models do, how their outputs arise, why fluent text is not factual authority, and what evidence institutions need before generated content influences decisions.

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LESSON ORIENTATION

Purpose and Reader Guidance

The central conclusion is simple: language generation is probabilistic prediction embedded in a larger sociotechnical system.

The lesson begins with transparent mechanisms—tokens, conditional probability, decoding and attention—then connects them to retrieval, evaluation, governance and accountable use.

Mathematical notation is kept interpretable, and each technical idea is paired with an operational consequence.

Readers do not need to train a large model. The companion notebook uses small deterministic examples to reveal mechanisms that are otherwise hidden inside large systems. The examples support reasoning; they do not claim to reproduce frontier-scale performance.

Concept	Operational meaning
Technical pathway	Study mechanisms, equations, code experiments and evaluation measures.
Executive pathway	Focus on decision boundaries, evidence, risk ownership and authorization.
Shared outcome	Recognize when an output is useful, unsupported, unsafe or unfit for a decision.

How to use the lesson: Read sections 1–12 in order, run the notebook laboratories, complete the exercises, and use the assessment to test independent understanding.

COMPETENCY MAP**Learning Objectives**

- Distinguish generative, discriminative and representation-learning objectives.
- Explain tokenization, embeddings, context and next-token prediction.
- Interpret autoregressive factorization and scaled dot-product attention.
- Compare deterministic and stochastic decoding policies.
- Separate parametric memory from retrieved evidence and tool results.
- Evaluate fluency, groundedness, correctness, robustness, safety and decision fitness.
- Specify an auditable evidence chain for institutional deployment.

Competency	Observable performance
Explain	Describe mechanisms accurately without anthropomorphism.
Demonstrate	Run transparent experiments and interpret outputs.
Evaluate	Test claims against authorized evidence and use-case requirements.
Govern	Assign controls, records, escalation and accountable approval.

MOTIVATING CASE**The Brief That Sounded Certain**

A public-enterprise analyst asks a language model to summarize a quarterly report. The response is fluent, concise and wrong about one financial figure. It also silently converts a provisional estimate into an audited result. Because the prose sounds confident, the error travels into an executive briefing.

The failure cannot be reduced to one bad number. The workflow did not preserve source authority, provisional status, model settings, claim-level verification or accountable approval. A better prompt might reduce error, but it cannot replace the evidence and control system.

Concept	Operational meaning
Observed symptom	Confident but unsupported financial statement
Technical cause	Generation optimized plausible continuation, not source fidelity
Process cause	No claim-to-source verification before reuse
Decision risk	Management acts on a materially altered statement
Required response	Trace, correct, document, assess impact and strengthen controls

Reasoning question: If the final number happens to be correct, what evidence would still be missing?

FOUNDATIONS

Generative and Discriminative Learning

A discriminative model estimates an outcome or label from observed features. A generative model learns a distribution over data, or a conditional distribution, from which new samples can be produced. The distinction concerns the modelling objective, not whether one system is intelligent and another is not.

$$\text{Discriminative: } p(y | x) \qquad \text{Generative: } p(x) \vee p(x | c)$$

Concept	Operational meaning
Classification	Estimate the probability of a label from features.
Regression	Estimate a numeric outcome or conditional expectation.
Unconditional generation	Produce samples from a learned data distribution.
Conditional generation	Produce samples given a prompt, class, image, instruction or other context.

Modern applications often combine these functions. A language model may generate text, classify intent through prompting, retrieve documents through an external component and call a deterministic tool. System analysis must identify which component produced each material claim.

FOUNDATIONS

What Generative AI Can Produce

Generative AI covers systems that produce derived synthetic content such as text, images, audio, video, code and structured records. A foundation model is trained broadly and then adapted or instructed for many downstream tasks. Not every generative model is a large language model, and not every AI system generates content.

Concept	Operational meaning
Text	Drafting, summarization, translation, extraction and dialogue
Code	Completion, transformation, tests and explanation
Image and video	Creation, editing, synthesis and multimodal reasoning
Audio	Speech generation, transcription support and transformation
Structured output	JSON, tables, classifications and workflow actions

Capability descriptions must include conditions. Performance depends on language, domain, task definition, model version, retained context, tools, decoding policy and evaluation method. A benchmark score is not a universal certificate for every population or workflow.

REPRESENTATION

From Text to Tokens

Models operate on token identifiers rather than directly on words or human meaning. A tokenizer segments input according to a fixed vocabulary and algorithm. The same sentence can consume different numbers of tokens under different tokenizers, particularly across languages, scripts, names and technical terminology.

Concept	Operational meaning
Word tokens	Intuitive but brittle for rare forms and multilingual text
Character tokens	Broad coverage but long sequences
Subword tokens	Reusable fragments balancing coverage and sequence length
Byte-level tokens	Very broad coverage with potentially uneven efficiency

Tokenization affects context capacity, latency, cost and representational equity. Institutions should measure actual documents in the languages they serve. English-only estimates are insufficient for a multilingual public service.

Notebook link: Part A constructs a visible tokenizer, vocabulary and encoding process, then examines a French agricultural-statistics sentence.

REPRESENTATION

Embeddings and Contextual Representation

A token identifier is an index, not a semantic measurement. An embedding table maps identifiers to vectors. Transformer layers then update token representations by combining information from the retained sequence. The resulting vector is contextual: the representation of a word can change with surrounding text.

Concept	Operational meaning
Identifier	Integer referring to one tokenizer entry
Embedding	Learned vector associated with an identifier
Position	Information representing token order
Contextual state	Layer output after information has been mixed across positions
Logits	Unnormalized scores for possible next tokens

Vector proximity can encode useful statistical regularities, but it does not automatically establish truth, fairness or causal meaning. Embeddings may reproduce patterns and exclusions present in training data.

REPRESENTATION

The Context Window as an Information Boundary

The context window contains the tokens available for the current computation: system and developer instructions, conversation history, retrieved passages, user input and reserved output. Material outside this boundary cannot directly support the current generation unless another component retrieves or reintroduces it.

$$n_instructions + n_history + n_sources + n_input + n_output\ reserve \leq C$$

A longer context window increases capacity but does not guarantee that every retained statement will influence the answer correctly. Position, redundancy, conflicts, prompt structure and model behaviour still matter.

Concept	Operational meaning
Include	Authorized, relevant and current evidence
Exclude	Sensitive, irrelevant, prohibited or stale content
Reserve	Enough tokens for the required response
Disclose	Material evidence omitted through truncation or retrieval limits

GENERATION

Autoregressive Sequence Probability

An autoregressive language model represents a sequence as a product of conditional next-token probabilities. At generation time, it repeatedly predicts a distribution, chooses one token, appends it to context and calculates the next distribution.

$$p(x_1, \dots, x_T) = \prod_{t=1}^T p(x_t | x_1, \dots, x_{t-1})$$

This factorization explains both power and limitation. Complex paragraphs emerge from many local conditional steps, but the objective does not contain a built-in guarantee that a sentence is current, supported or institutionally authorized.

Interpretation boundary: High probability means compatibility with the learned model and context. It does not mean factual truth.

Concept	Operational meaning
Likelihood	Probability assigned by the model
Factuality	Agreement with verified reality
Groundedness	Support from identified authorized evidence
Decision fitness	Suitability for an accountable workflow

MODEL LIFECYCLE**Pretraining and Learned Parameters**

During pretraining, a model updates parameters to improve a predictive objective over large datasets. Parameters compress statistical structure; they are not a transparent catalogue of source documents. The model may reproduce patterns, combine fragments, generalize, omit qualifications or generate unsupported continuations.

- Define data sources, rights and exclusions.
- Filter and document quality and safety decisions.
- Specify objective, architecture and optimization controls.
- Measure performance across relevant languages and subgroups.
- Retain lineage, version and evaluation records.

A model's internal knowledge can be incomplete, outdated or conflicting. For current institutional claims, training-time exposure is not a sufficient provenance mechanism.

MODEL LIFECYCLE

Instruction Tuning Alignment and Adaptation

Pretrained models are commonly adapted to follow instructions and preferred behaviours. Supervised instruction tuning uses demonstrations. Preference-based methods compare or score outputs. Parameter-efficient methods update a small subset of parameters or adapters. Prompting and retrieval can change behaviour without parameter updates.

Concept	Operational meaning
Prompting	Changes current instructions and context; no parameter update
Retrieval	Adds external evidence to the current context
Fine-tuning	Updates model parameters using task or domain data
Preference optimization	Shapes behaviour using comparative or reward signals
Tool integration	Delegates actions or calculations to external systems

Each method creates different evidence and risks. A workflow must not describe retrieval as training, tool output as model knowledge, or prompt examples as parameter updates.

TRANSFORMERS

Why Attention Matters

The Transformer architecture replaced recurrence with attention-based sequence processing, enabling parallel computation during training and flexible interaction among token positions. Contemporary language models extend and modify this foundation, but scaled dot-product attention remains a central conceptual building block.

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

Concept	Operational meaning
Query	What the current position seeks
Key	How another position can be matched
Value	Information contributed after weighting
Scale	Controls score magnitude using key dimension
Softmax	Converts scores into row-wise weights

Notebook link: Part D calculates attention with three small vectors and displays the resulting weight matrix.

TRANSFORMERS

Attention Is Not a Complete Explanation

Attention weights describe one internal mixing operation. They do not automatically identify causal reasoning, source provenance or the human meaning of a decision. Multiple layers, heads, residual connections, nonlinear transformations and decoding steps contribute to the final output.

Concept	Operational meaning
Valid observation	A position assigned a particular weight in a defined layer and head
Invalid leap	The weight proves why the system reached its final conclusion
Required evidence	Controlled interventions, evaluation, traces and external verification
Communication rule	Describe measured internal behaviour without anthropomorphic certainty

Interpretability evidence should be matched to the decision. Debugging a model component, explaining an administrative action and proving factual support require different methods.

INFERENCE

Prompts Instructions and Retained Context

A prompt is not merely the user's final question. Applications often assemble system rules, developer instructions, history, retrieved evidence, templates and tool descriptions. The assembled request forms an information and authority boundary for the current inference.

- State the task and intended audience.
- Provide authorized context and preserve source labels.
- Define output structure and uncertainty requirements.
- Specify prohibited actions and escalation conditions.
- Require claim-level support when facts matter.

Prompt quality matters, but prompts cannot guarantee compliance or factuality. Treat them as one control within a layered system of access restrictions, retrieval policy, validation, monitoring and human authorization.

INFERENCE

Greedy and Stochastic Decoding

Greedy decoding selects the highest-probability token at every step. Sampling draws from a probability distribution. Beam search retains several partial candidates. No decoding method converts probability into factual authority; each manages trade-offs among determinism, diversity, repetition and computational cost.

Concept	Operational meaning
Greedy	Repeatable for fixed model and context; may be rigid or locally suboptimal
Temperature	Flattens or sharpens the distribution
Top-k	Samples from the k highest-probability candidates
Top-p	Samples from the smallest set reaching cumulative probability p
Beam search	Tracks several high-scoring partial sequences

Model version, prompt, context and decoding settings belong in the operational record when outputs are material.

INFERENCE

Temperature Top K and Top P

Temperature divides logits before softmax. Values below one concentrate probability mass; values above one flatten it. Top-k fixes the number of candidates, while top-p adapts the candidate count to the shape of the distribution.

$$P_T(i) = \exp(z_i / T) / \sum_j \exp(z_j / T)$$

- Use lower variability when consistency matters.
- Test rather than assume the safest setting.
- Evaluate rare but material failures across repeated runs.
- Record settings and random seeds in experiments.
- Do not treat temperature zero as a universal reproducibility guarantee across services or versions.

Practice: The notebook compares distributions at three temperatures and samples with greedy, top-k and top-p policies.

CAPABILITY**Useful Functions and Conditional Performance**

Language models can summarize, transform, classify, translate, extract, draft and support software work. Performance is conditional on the task, language, domain, evidence, model version, context construction and evaluation design.

Concept	Operational meaning
Summarization	Preserve material facts, qualifications and source boundaries
Extraction	Validate schema, omissions and transcription accuracy
Drafting	Separate generated wording from approved institutional positions
Classification	Measure error by class and affected subgroup
Code support	Test correctness, security and license implications

A successful demonstration is evidence about one defined setting. It is not authorization for another population, dataset or consequential workflow.

LIMITATIONS**Confabulation and Unsupported Specificity**

A model can generate a plausible statement not supported by the available evidence. NIST uses confabulation to describe confidently stated but erroneous or false content. The term keeps attention on observable output rather than implying a human mental state.

Concept	Operational meaning
Fabricated citation	Reference does not exist or does not support the claim
Altered number	Value, unit, period or status changes
False completion	Missing evidence is filled with a plausible detail
Source blending	Claims from different documents are combined incorrectly
Overclaim	Uncertainty or limitation disappears in the summary

Controls include retrieval from authorized sources, claim decomposition, citations, deterministic calculations, review thresholds and refusal or escalation when evidence is insufficient.

LIMITATIONS

Output Agreement Does Not Prove Reasoning

A correct answer on one case does not reveal the internal route, reliability under variation or behaviour on the next case. The model may exploit shortcuts, memorize patterns, infer correctly or arrive at the answer through an unstable sequence of steps.

- Vary surface form while preserving meaning.
- Test counterexamples and boundary cases.
- Separate final-answer scoring from evidence and process checks.
- Compare repeated runs when sampling is enabled.
- Use external tools for calculations that require exactness.

Thinking box: What additional evidence would you require if the output affected payment, eligibility, official publication or safety?

GROUNDING

Parametric Memory and Retrieval

Parametric memory is encoded in learned weights. Retrieval introduces external records at inference time. Retrieval-augmented generation can improve specificity, updateability and provenance, but only if retrieval finds authoritative passages and the generator uses them faithfully.

Concept	Operational meaning
Indexing	Represent and organize eligible sources
Retrieval	Select passages for the current request
Reranking	Reorder candidates using stronger relevance evidence
Generation	Produce an answer conditioned on retained passages
Verification	Check each material claim against its cited support

Retrieval failure and generation failure must be measured separately. A perfect answer is impossible when the decisive source was never retrieved, but retrieved evidence can also be ignored or contradicted.

GROUNDING

Claim Level Evidence and Citation Discipline

Groundedness asks whether material claims are supported by identified evidence. A document-level citation may be insufficient when one sentence combines several values, periods or qualifications. Claim-level tracing makes verification and correction faster.

Concept	Operational meaning
Claim	Quarterly revenue was USD 12.4 million
Source span	Authorized table cell and associated note
Qualifier	Provisional pending audit
Verification	Value, unit, period and status all match
Disposition	Approve, revise, reject or escalate

A citation is not proof by itself. The cited passage must exist, be authoritative for the claim, and support the meaning attributed to it.

SYSTEMS

Tools Functions and Deterministic Computation

An application may allow a model to call a database, calculator, search service or business function. The model proposes a structured action; the surrounding system validates permissions and executes it. Tool integration can improve accuracy but creates security and authorization risks.

- Validate arguments against a strict schema.
- Apply least-privilege access independently of model text.
- Require confirmation for consequential or irreversible actions.
- Treat retrieved content as untrusted input.
- Log tool requests, results, errors and human approvals.

Never assume that because a model can describe an action it is authorized to perform it.

SYSTEMS

Agentic Loops and Control Boundaries

Agentic systems iterate among planning, tool use, observation and revision. Each additional step can improve task completion while expanding the opportunity for compounding error, prompt injection, excessive access and unclear responsibility.

Concept	Operational meaning
Plan	Decompose a goal within allowed scope
Act	Invoke an authorized tool with validated arguments
Observe	Receive and classify results or errors
Revise	Update the plan without silently expanding authority
Stop	Complete, refuse or escalate using explicit criteria

Bound autonomy by budget, time, tools, data, side effects and approval gates. The accountable institution—not the model—owns the decision to deploy and the consequences of operation.

EVALUATION**Separate the Dimensions That Matter**

Dimension	Evaluation question
Fluency	Coherence, readability and relevance
Groundedness	Support from authorized evidence
Correctness	Agreement with verified reality
Robustness	Stability across valid variation and perturbation
Safety	Avoidance of prohibited or harmful behaviour
Decision fitness	Suitability for the actual accountable workflow

A single composite score can hide a material failure. Report dimensions separately, define thresholds in advance and preserve failure examples. Evaluation should reflect the intended languages, populations, document types and consequences.

Key distinction: A fluent answer can be wrong; a grounded answer can be incomplete; a correct answer can still be unauthorized.

EVALUATION

Designing Credible Evaluation

- Define intended and excluded uses before selecting metrics.
- Build representative and challenge sets without test contamination.
- Preserve reference answers, source evidence and scoring rubrics.
- Use qualified human review where judgment is necessary.
- Measure subgroup and language performance when impacts differ.
- Record model, prompt, retrieval and decoding versions.
- Retest after material system changes.

Evaluation evidence has an information boundary. Results support claims only for the tested configuration and distribution. Moving from a pilot to production changes users, incentives, data and failure consequences.

Concept	Operational meaning
Offline	Controlled datasets, scenarios and red-team cases
Pilot	Limited users and monitored workflows
Production	Live performance, incidents, drift and human outcomes

EVALUATION

Human Review Without False Assurance

Human review is effective only when reviewers have time, authority, evidence and a clear rubric. Asking a person to approve large volumes of plausible text can create automation bias and rubber-stamping.

- Display source evidence beside generated claims.
- Highlight uncertainty and missing evidence.
- Prioritize high-impact differences rather than cosmetic edits.
- Allow rejection and escalation without productivity penalties.
- Sample approved outputs for independent quality assurance.
- Measure reviewer agreement and correction rates.

Operational test: If a reviewer cannot reconstruct why a claim was approved, the workflow is not adequately auditable.

RISK AND CONTROL

Privacy Confidentiality and Data Minimization

Prompts, attachments, retrieved passages, outputs and logs can contain personal, commercial or security-sensitive information. Data protection must cover the complete application path, including vendors, plugins, caches, monitoring systems and human review queues.

Concept	Operational meaning
Minimize	Send only information necessary for the defined task
Authorize	Enforce purpose, role and source permissions
Protect	Apply encryption, isolation and secret management
Retain	Define justified periods for prompts, outputs and traces
Delete	Verify deletion obligations across components and providers

Do not paste sensitive records into an unapproved service merely because the interface is convenient.

RISK AND CONTROL**Prompt Injection and Untrusted Content**

Retrieved documents and webpages may contain instructions designed to redirect the model, disclose information or misuse tools. Because models process instructions and data through the same representational channel, application controls must distinguish authority outside the model.

- Treat external content as data, not instructions.
- Separate trusted policy from untrusted retrieved text.
- Restrict tools and data through conventional access controls.
- Validate outputs before they enter commands, queries or messages.
- Test indirect prompt injection and data-exfiltration scenarios.
- Escalate anomalous requests and tool sequences.

RISK AND CONTROL**Bias Language and Unequal Performance**

Training data, tokenization, evaluation coverage and deployment context can produce unequal performance. Lower-resource languages, local names and specialized institutional terminology may fragment differently or be underrepresented in evaluation sets.

Concept	Operational meaning
Representation	Who and which languages appear in data
Measurement	Whether metrics capture relevant harms
Allocation	Who receives benefits, burdens or errors
Feedback	How deployed outcomes alter future data
Recourse	How affected people challenge or correct results

Test the actual populations and languages served. Aggregate accuracy can conceal severe failure for a smaller group.

RISK AND CONTROL**Rights Provenance and Content Integrity**

Generative workflows raise questions about training-data rights, licensed inputs, confidential material, attribution and generated similarity. Institutions need a rights-aware process appropriate to jurisdiction, contract and use case.

- Document source and license conditions for supplied material.
- Avoid requesting long reproductions of protected content.
- Review generated outputs for inappropriate copying or disclosure.
- Preserve attribution when source material requires it.
- Define approval routes for public release and reuse.

Synthetic content should not be presented as an observed fact, authentic record or verified quotation without appropriate disclosure and validation.

OPERATIONS

Latency Cost and Environmental Resources

Model choice, prompt length, output length, retrieval, tool use and repeated retries affect latency and cost. Larger is not automatically better. A smaller model or deterministic method may meet the requirement with lower risk and resource demand.

Concept	Operational meaning
Task routing	Use the simplest capable method
Context control	Remove duplication while preserving necessary evidence
Caching	Reuse approved stable results with version controls
Batching	Group suitable workloads without mixing authorization contexts
Monitoring	Track cost per accepted output, not only cost per call

Decision rule: Compare quality, risk, latency and total operating cost against a credible non-LLM baseline.

OPERATIONS**Observability Incidents and Change Control**

Production monitoring should connect technical events to user and institutional outcomes. Model-provider changes, prompt revisions, retrieval updates and tool modifications can alter behaviour even when the user interface appears unchanged.

- Version prompts, models, indexes, policies and tool schemas.
- Monitor refusals, unsupported claims, latency, cost and escalation.
- Retain representative failure cases subject to privacy limits.
- Define incident severity, ownership and notification routes.
- Revalidate after material configuration or data changes.
- Maintain rollback or service-disable procedures.

SRAI CONTROL

The Generative AI Evidence Chain

For every consequential output, connect the request to authorized context, the exact system configuration, the unedited generated output, verification evidence and the accountable human decision.

Concept	Operational meaning
Request	Task, user, purpose and intended audience
Authorized context	Sources, versions, permissions and retrieval record
System	Model, prompt, tools and decoding settings
Output	Original response and structured actions
Verification	Claim checks, exceptions and reviewer evidence
Decision	Approval, revision, rejection or escalation

Retain enough information to reproduce and audit the chain without retaining more sensitive information than necessary.

SRAI CONTROL

SRAI Principle Number 96

A language model generates statistically plausible continuations; factual authority must come from evidence, validation and accountable human use.

The principle does not deny model capability. It locates authority correctly. Models can transform and synthesize information, but institutional claims require sources, controls and responsibility that exist outside the probability distribution.

Concept	Operational meaning
Plausibility	A property of generation
Evidence	Support for a claim
Validation	Testing against defined requirements
Authority	Legitimate institutional basis for action
Accountability	Human and organizational responsibility

APPLIED CASE

Official Statistics Narrative Generation

An agricultural-statistics office uses an LLM to draft commentary from validated tables. A defensible workflow preserves indicator definition, unit, geography, reference period, population, revision status, uncertainty and disclosure rules.

Concept	Operational meaning
Input	Approved table, metadata and disclosure status
Generation	Draft narrative with source identifiers
Validation	Numeric, semantic and disclosure checks
Review	Subject-matter statistician compares claims to evidence
Release	Authorized publication with version record

A correct number can still mislead if detached from its unit, denominator, population, period or provisional status. For official statistics, metadata is part of the fact.

Control preference: Use deterministic code to calculate values and changes; use the model to draft language around verified results.

APPLIED CASE**Public Enterprise Quarterly Reporting**

A reporting system receives structured financial data from public enterprises through database exchange, scheduled files or controlled templates. The LLM may explain exceptions and draft summaries, but source systems remain authoritative for amounts and statuses.

- Retrieve only the approved reporting period and entity.
- Calculate ratios and totals in deterministic services.
- Preserve currency, scale, consolidation and audit status.
- Require citations for material narrative claims.
- Route anomalies and missing submissions to named reviewers.
- Prevent the model from directly posting or altering ledger records.

This separation allows useful language assistance without confusing a generative interface with the financial system of record.

NOTEBOOK LABORATORY

Practical Work and Interpretation

The companion notebook contains 41 cells: 21 executable code cells and 20 instructional cells. It runs independently in VS Code and Google Colab and does not require private SRAI packages or external APIs.

Concept	Operational meaning
Part A	Tokenizer, vocabulary, encoding and multilingual behaviour
Part B	Bigram next-token probabilities and sequence log probability
Part C	Temperature, greedy, top-k and top-p decoding
Part D	Transparent NumPy self-attention
Part E	Grounding and multidimensional evaluation
Part F	Evidence chain and decision-fitness controls

Interpret results at the level the demonstration supports. The notebook reveals mechanisms; it does not validate a production model or institutional application.

PRACTICE**Exercises**

1. Contrast discriminative conditional probability with autoregressive sequence probability.
2. Trace one sentence through tokenization, identifiers, embeddings and next-token logits.
3. Explain why temperature changes variability but not factual authority.
4. Design a context budget for a multilingual quarterly report.
5. Create a claim-to-source verification table for an official statistic.
6. Identify three prompt-injection paths in a retrieval-and-tool application.
7. Separate retrieval failure from generation failure in an evaluation design.
8. Define an escalation rule for an unsupported high-impact claim.
9. Specify the records needed to reproduce a consequential output.
10. Compare an LLM workflow with a credible deterministic baseline.

Submission expectation: For each answer, state the claim, supporting reasoning, operational consequence and evidence required.

ASSESSMENT**Student Assessment**

1. What distinguishes a generative model from a discriminative model?
2. What does an autoregressive language model estimate at each step?
3. Why must training, adaptation, inference, retrieval and tool use be distinguished?
4. Which dimensions should be evaluated before consequential use?
5. Which provenance and governance information should accompany a deployed LLM application?

Each question is worth 4 marks. Award 4 for a correct result with valid reasoning and operational interpretation; 3 for a correct core result with limited interpretation; 2 for partial structure; 1 for a relevant attempt; and 0 for no relevant evidence.

Criterion	Evidence of mastery
Conceptual accuracy	Uses terms and mechanisms correctly
Reasoning	Connects claims to evidence
Application	Identifies operational consequences
Boundary	States limitations and conditions

ASSESSMENT

Instructor Solutions and Marking Guidance

1. Answer: A generative model learns a distribution from which new outputs can be produced; a discriminative model estimates outcomes or labels conditional on features.

Marks: 4 correct and well reasoned; 3 substantially correct; 2 partial structure; 1 relevant attempt.

2. Answer: It estimates a conditional probability distribution over the next token given the retained context.

Marks: 4 correct and well reasoned; 3 substantially correct; 2 partial structure; 1 relevant attempt.

3. Answer: They introduce different data flows, parameter changes, permissions, failure modes and evidence requirements.

Marks: 4 correct and well reasoned; 3 substantially correct; 2 partial structure; 1 relevant attempt.

4. Answer: Evaluate fluency, groundedness, correctness, robustness, safety, subgroup behaviour, latency, cost and decision fitness.

Marks: 4 correct and well reasoned; 3 substantially correct; 2 partial structure; 1 relevant attempt.

5. Answer: Record model and data provenance, intended and excluded uses, evaluation sets, prompt and context policy, decoding settings, safeguards, limitations, residual risks and accountable approval.

Marks: 4 correct and well reasoned; 3 substantially correct; 2 partial structure; 1 relevant attempt.

CLOSING SYNTHESIS**Key Insights Glossary and References**

- Generation is repeated conditional prediction.
- Fluency is not factual authority.
- Context and decoding are part of the system configuration.
- Retrieval can supply evidence but does not guarantee faithful use.
- Evaluation must match the intended workflow and affected population.
- Accountable institutions retain responsibility for deployment and decisions.

Key terms: autoregressive model; attention; confabulation; context window; decoding; embedding; foundation model; groundedness; inference; retrieval-augmented generation; token; transformer.

Selected references

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Transition: Lesson 2 examines tokenization and context windows as measurable operational boundaries.